

Various notes on MLS Chapter 2, Sections 2 & 3

On eq. (2.3) p. 25:

A condition for using equation (2.3) that is not mentioned by MLS is that the rotation matrix R_{ab} must be expressed in components in frame A and the rotation matrix R_{bc} must be expressed in components in frame B . Even though this is not mentioned, it is implied by the way that the rotation matrix is defined on p. 23.

On eq. (2.22) p. 35:

A condition for using equation (2.22) that is not mentioned by MLS but that follows from the interpretation of what is stated in the equation is that the position vector p_{ab} must be expressed in frame A .

On eq. (2.24) p. 37:

Two conditions for using equation (2.24) that are not mentioned by MLS are that

- 1.) the rotation matrix R_{ab} must be expressed in components in frame A and the rotation matrix R_{bc} must be expressed in components in frame B .
- 2.) the position vector p_{bc} must be expressed in the frame B and the position vector p_{ab} in frame A .

Even though these conditions are not mentioned, it is implied by the way that the rotation matrix is defined on p. 23 and the position vector is defined on p. 35.

Proof of the identity, eq. (2.35) p. 42:

In eq. (2.34) we defined $\hat{\xi}' = g^{-1}\hat{\xi}g$. Therefore $\hat{\xi} = g\hat{\xi}'g^{-1}$. Therefore

$$\begin{aligned} e^{\hat{\xi}\theta} &= e^{g\hat{\xi}'g^{-1}\theta} = e^{g\hat{\xi}'\theta g^{-1}} \\ &= I + g\hat{\xi}'\theta g^{-1} + \frac{(g\hat{\xi}'\theta g^{-1})^2}{2!} + \frac{(g\hat{\xi}'\theta g^{-1})^3}{3!} \dots \end{aligned}$$

But, for instance,

$$\frac{(g\hat{\xi}'\theta g^{-1})^3}{3!} = \frac{1}{3!}g\hat{\xi}'\theta g^{-1}g\hat{\xi}'\theta g^{-1}g\hat{\xi}'\theta g^{-1} = \frac{1}{3!}g\left(\hat{\xi}'\theta\right)^3g^{-1}$$

Therefore,

$$\begin{aligned} e^{\hat{\xi}\theta} &= I + g\hat{\xi}'\theta g^{-1} + g\frac{(\hat{\xi}'\theta)^2}{2!}g^{-1} + g\frac{(\hat{\xi}'\theta)^3}{3!}g^{-1} \dots \\ &= g\left(I + \hat{\xi}'\theta + \frac{(\hat{\xi}'\theta)^2}{2!} + \frac{(\hat{\xi}'\theta)^3}{3!} \dots\right)g^{-1} \\ &= ge^{\hat{\xi}'\theta}g^{-1} \end{aligned}$$

On the definition of the pitch of a screw, bottom p. 45:

Note that in machine design (and in machine theory in general) the pitch of a physical single thread screw is defined as the translation d when the screw rotates through one revolution; that is, translation per unit revolution, $d/(2\pi)$. This is similar to, but not exactly the same as, how pitch is defined within the context of this course, in the last paragraph of p. 45, namely translation per unit angle, d/θ .

On the special case of a screw with zero pitch p. 47:

From Lemma 2.3 (p. 28) follows that $\hat{\omega}^2 = \omega\omega^T - \|\omega\|^2 I$ and $\hat{\omega}^3 = -\|\omega\|^2 \hat{\omega}$ (that is, without any limitation that $\|\omega\| = 1$). Consider a screw with a zero pitch $h = 0$, which corresponds with the motion associated with a revolute joint. If, as suggested in the first paragraph following the unnumbered equation after eq. (2.40) p. 47, we choose $v = -\hat{\omega}q + h\omega = -\hat{\omega}q$, the 2nd term in the element in the 1st row, 4th column, of $e^{\hat{\xi}\theta}$, $\omega\omega^T v\theta$, can be simplified:

$$\omega\omega^T v\theta = -\omega\omega^T \hat{\omega}q\theta = -(\|\omega\|^2 I + \hat{\omega}^2)\hat{\omega}q\theta = -(\|\omega\|^2 \hat{\omega} + \hat{\omega}^3)q\theta = -(\|\omega\|^2 \hat{\omega} - \|\omega\|^2 \hat{\omega})q\theta = 0.$$

Therefore, when $h = 0$,

$$e^{\hat{\xi}\theta} = \begin{bmatrix} e^{\hat{\omega}\theta} & (I - e^{\hat{\omega}\theta})\hat{\omega}v \\ 0 & 1 \end{bmatrix}.$$